



Atlas of Lunar Historical Sites and Proposed Bases and Infrastructure

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Abstract

The 21st century has witnessed an expansion of space operations throughout the Earth–Moon system and the wider Solar System with the introduction of new space-faring countries, decreasing costs of space access, and advances in space system technologies. Many countries are endeavoring to move outside geosynchronous orbit to pursue missions in cislunar space and lunar orbit, as well as on the lunar surface, with invigorated U.S., Chinese, and Russian lunar mission initiatives being principle examples. Coalescing international efforts to return to the Moon may result in not only the pursuance of short-duration crewed lunar missions, but also the attainment of the first extraterrestrial human settlement(s) and the establishment of a permanent lunar base. Site selection for short-duration soft landings and the construction of basing/infrastructure will become a critical function to enhance mission assurance and operational safety. In this paper, an atlas of human activity on the lunar surface is presented featuring a concise mapping of locations associated with historical lunar impactor and soft-landing sites and proposed future basing and infrastructure reports. The atlas seeks to enable a holistic depiction of lunar surface activity to gain insight into where future lunar bases may be located and establish historical patterns of proposed basing.

Keywords Atlas · Lunar debris · Historical lunar sites · Lunar South pole · Astroarchaeology

1 Introduction

The 2010s and early 2020s have marked a shift in the Earth-centric space operations paradigm, with a renewal in international interest in pursuing missions extending into the cislunar regime, lunar orbit, and the lunar surface. Recent international activity includes, but is not limited to, China's Chang'e-4 far-side lunar mission in 2018, China's Chang'e-5 lunar sample-return in 2020, the first launch of the U.S. Artemis launch vehicle in 2022, and plans to develop a

joint Chinese-Russian base at the lunar South Pole in the 2036–2045 period [1–4]. The return of crewed missions to the Moon may evolve into permanent human settlements with the establishment of bases and associated infrastructure for functions such as power generation, communications, and food production/horticulture. Site selection for basing and infrastructure represents a critical first step to any lunar construction initiative, and future lunar missions will require the completion of specialized terrain surveying to update existing cartographical and geologic data. However, cartographical data associated with historical lunar missions and proposed basing has remained uncollected and scattered in disparate sources since the early Space Age, and a concise atlas of lunar surface activity has eluded compilation until now.

By definition, an atlas represents “a collection of maps often including illustrations, informative tables, or textual matter” [5]. This paper will provide a collection of maps and associated tables related to locations of interest on the lunar surface corresponding to historical lunar missions and a proposed lunar base. A standard set of lunar surface maps is used as the cartographical basis for this lunar basing atlas. While using a single map of the lunar surface was possible, the observed groupings of historical and

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proposed lunar mission activity necessitated the division of the lunar surface into three primary zones: North Polar region (90° N to 60° N latitude), Midlands region (60° N to 60° S latitude), and the South Polar region (60° S to 90° S latitude). A stereographic projection was employed to generate the North/South Polar maps [6], while a Mercator projection creates the Midlands region map [7].

This atlas is divided into two sections with respect to the type of human activity on the lunar surface. The first type of activity focuses on historical lunar landing and impact sites corresponding with crewed and uncrewed lunar missions covering the period from the early Space Age to the contemporary era. Historical sites are presented collectively, as well as independently with respect to the country-of-origin for the mission. The second type of activity relates to the planned construction of lunar bases and infrastructure. While each type of activity features its own collection of maps and associated tables, a combined map is also given to enable a holistic depiction of lunar surface activity to gain insight into where future lunar bases may be located, and to establish historical patterns of proposed basing. Current trends in space system and mission development for cislunar and lunar operations indicate a strong likelihood for the establishment of permanent human settlements on the Moon in the next few decades. As a result, this paper represents only the first edition of an atlas that will come to encompass the dynamic evolution of international lunar basing activity. As of late 2022, the potential landing sites for Artemis III have been announced [8]. These sites, although mostly in the South Polar region, are integrated into this atlas, along with the attractive qualities of the South Polar region, and represents only the latest in a long line of landing site and basing plans associated with lunar surface exploration.

It is worthwhile here to include some definitions that may help the reader understand the text due to words having a variety of definitions when compared to day-to-day use. These generic definitions are given and will be seen generously throughout this atlas.

1. **Land (or landing):** The action of when a vehicle reaches an altitude of zero with respect to its predetermined surface of arrival. This action is intentional and not the result of a vehicle failure or disposal action.
2. **Impact:** An intentional collision with the lunar surface for a mission-defined purpose.
3. **Crash:** An intentional (i.e., disposal) or non-intentional (i.e., uncontrolled impact) end-of-life action of a vehicle colliding with the lunar surface.
4. **Site:** A region/location where a vehicle has purposely and/or unintentionally arrived at for a specific mission/purpose. A typical site contains information regarding what country is responsible, location, and date.

5. **Basing:** The action of establishing a location to build a facility and/or infrastructure of interest.

Overall, this atlas expands existing literature with respect to lunar surface activity from the perspective of historical and contemporary missions, proposed basing locations, and anticipated locations for future landing locations and exploration. This atlas information and analysis related to the following objectives:

1. Compiles and presents a singular repository of historical data regarding all lunar missions and lunar surface activity starting at the dawn of the Space Age, to include vehicle soft landings, impacts, and crashes. This mission and surface activity data is depicted in both graphical and tabular format.
2. Identifies propositions for lunar surface locations to use with basing and infrastructure.
3. Examines the relative locations of combined historical sites and proposed basing locations.
4. Explores risks associated to historical and proposed lunar sites due to debris impacts associated with debris propagation and dynamics studies for low lunar orbit.

2 Historical Lunar Sites and Astroarchaeology

A starting point for documenting sites of interest for a lunar atlas is recording historical sites or astroarchaeology. The term *astroarchaeology* (sometimes referred to as *archeo-astronomy*) has traditionally been used in conjunction with the archaeological research of ancient civilizations and their study of astronomical phenomena [9, 10] and the investigation of the stellar origins of astronomical bodies such as stars and galaxies [11, 12]. Recently, it has been applied to the archaeological study of human space activity, such as the investigation of the celestial whereabouts of the Eagle lunar ascent module [13]. The inclusion of historical lunar missions into the atlas gives context to what areas were previously identified as viable locations for mission activity, and will give insight into how sites of interest may have changed over time when compared with planned future sites.

To narrow the scope with respect to countries of interest, this paper will present historical sites in which lunar landers (crewed or uncrewed) from various space-faring countries have either hard (i.e., impact) or soft-landed the lunar surface. The countries of interest for this map are the U.S., the Union of Socialist Soviet Republics (U.S.S.R.), the European Union (E.U.), the People's Republic of China (herein referred to as China), India, Israel, and Japan. Overall, these historical locations are especially of interest as the country-of-origin for the mission or the general international

community may elect to preserve them in the future as human presence on the Moon increases. Reasons for historical site preservation relate to a historic “first” for a country or humanity, such as the location of the first lunar landing of a human-built object (e.g., Luna 2), the first crewed lunar landing (e.g., Apollo 11), and the inaugural lunar mission of an emergent space program (e.g., Israel, India).

2.1 U.S. Historical Sites

The U.S. represents the largest historical landing site contributor to this atlas. During the 1960s, the U.S. initiated three programs that focused on remote sensing and imaging from lunar orbit, and the physical study of the lunar surface either through kinetic impact or soft-landing to preserve the instruments and enable long-duration data collection and downlink. These missions occurred between the years 1962 and 1972 [14, 15], and provide 18 distinct landing locations for this atlas. The first of these programs, Ranger (1961–1965), was designed to obtain high-resolution images of the lunar surface, with most intended to initiate a de-orbit maneuver to impact the Moon. Four Ranger missions (Ranger 1–3, and Ranger 5) never impacted the lunar surface and remained in lunar orbit in accordance with the specific mission profiles; all remaining missions (Ranger 4, and Ranger 6–9) did successfully impact the Moon [16].

The next lunar exploration program was the Surveyor space vehicle. This program, which was active from 1966 to 1968 and led up to the first Apollo launches, proved the feasibility of uncrewed soft-landings and collected relevant photographs and data to support the subsequent Apollo missions. All seven missions impacted the lunar surface with only Surveyor 2 and Surveyor 4 failing to soft-land on the surface [17]. Landing locations for five Ranger and all seven Surveyor missions are outlined in Table 1 and shown in Fig. 1 [14, 18–21]. The ostensible final program of the early Space Age, Apollo, consisted of seven planned and six executed crewed lunar landing missions. The first successful landing occurred on 20 July 1969 with the Apollo 11 mission; only three years later, Apollo 17 would become the last crewed mission on 19 December 1972 [22]. The six Apollo missions that landed on the lunar surface (excepting Apollo 13) and their associated landing locations are given in Table 1 and Fig. 1 [23]. Of note, the Apollo 12 landing site is nearly co-located with that of Surveyor 3. In November 1969, the lunar module Intrepid landed within 200 m of the Surveyor 3 vehicle in the Oceanus Procellarum, thereby demonstrating the capability of determining the precise location of an object (only a few meters in size) on the Moon and executing a pinpoint landing near said object [24].

Nearly three decades after the conclusion of the Apollo program, the U.S. launched two missions to image and analyze the lunar surface in support of a future return of crewed

missions to the Moon. The Lunar Prospector and L-CROSS' Centaur Probe successfully impacted the surface in 1999 and 2009, respectfully. Both performed successfully with the Lunar Prospector even reporting data indicating ice being present in regolith at the lunar poles, marking the first evidence of water being present at the lunar poles [25]. Impact locations for both of these missions are included in this atlas [25–28]. The latest U.S. mission to impact the lunar surface was the Lunar Atmosphere and Dust Environment Explorer (LADEE). Launched in 2013 to study the Moon's exosphere and dust environment, the orbiter executed a de-orbit maneuver in 2014 and purposely landed near the rim of the Sundman V crater [29]. The next installation of U.S. lunar surface activity is associated with Artemis III and following missions with proposed landing sites at the lunar South Pole [8].

2.1.1 Russian/Soviet Historical Sites

Being the sole competitor with the U.S. during the Space Race to the Moon in the 1960s, it is only natural for the U.S.S.R. to boast the second-highest number of historical lunar landings with 14 sites. The Luna program is the generic title given to the series of uncrewed lunar missions launched between 1959 and 1976 by the U.S.S.R. to collect information on the Moon for general scientific purposes, as well as to help plan future crewed landings [30]. A part of these landing missions were two uncrewed rovers, Lunokhod-1 and Lunokhod-2, which were deployed by Luna 17 (in 1970) and Luna 21 (in 1973), respectfully [31]. Table 1 outlines the landing site details for the 12 lunar impactor and lander Luna missions, and the two associated Lunokhod rovers. The coordinates given for the Lunokhod rovers represent only an approximation for the location of these robotic vehicles. During their successful missions, Lunokhod 1 and 2 travelled nearly 11 km and 37 km, respectively, from their Luna landers before vehicle failure [32]. The most-recent Russian lunar mission, Luna 25, resulted in failure when the lander crashed into the lunar surface in 2023 [33].

2.2 General International Historical Sites

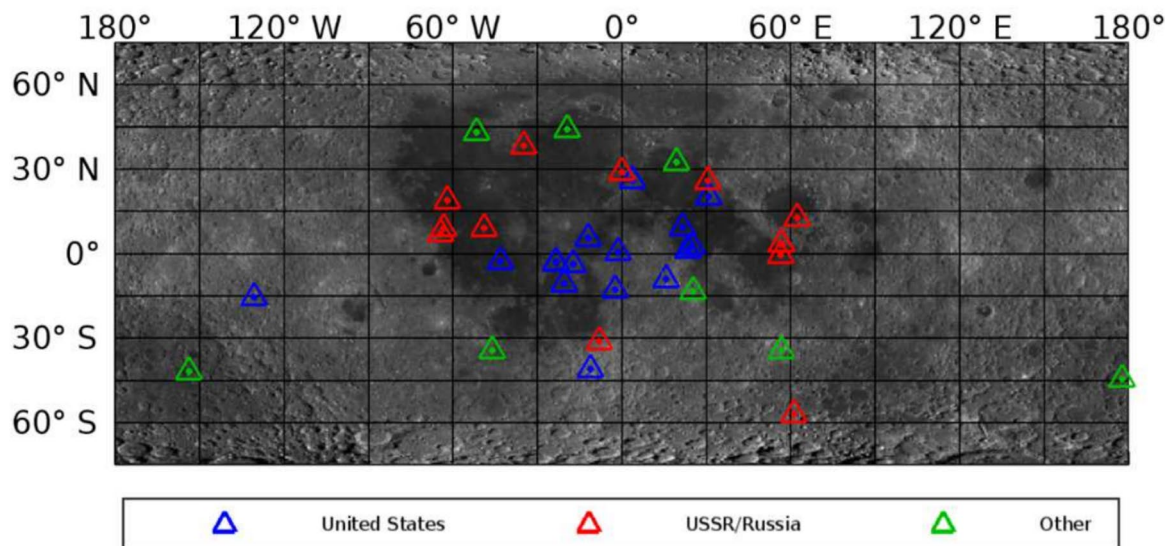
Historical landing sites from missions from countries other than the U.S. and U.S.S.R. include China, the E.U., India, Israel, and Japan. To start the list, three Chinese National Space Administration (CNSA) lunar missions are included corresponding to Chang'e 3, Chang'e 4, and Chang'e 5, a series of lunar soft-landing missions occurring between 2013 and 2020 [34], along with their most recent mission Chang'e 6 [35]. It is worth noting that Chang'e 4 landed on the far side of the Moon, thus making it the first and only soft-landing on the lunar far-side [33]. With Chang'e 4 intended to survive the landing, the system was accompanied by the Queqiao communications relay satellite. This relay

Table 1 Historical lunar sites sorted by country and year

Mission name	Latitude	Longitude	Country	Purpose	Year
Chang'e 3 [45]	44.1° N	19.5° W	China	Lunar Lander	2013
Chang'e 4 [46]	45.5° S	177.6° E	China	Lunar Lander	2019
Chang'e 5 [47]	43.1° N	51.8° W	China	Lunar Lander	2020
Chang'e 6 [35]	41.6° N	153° W	China	Lunar Lander	2024
SMART-1 [36]	34.3° S	46.2° W	E.U	Lunar Impactor	2006
Hiten [42]	34.3° S	56.6° E	Japan	Lunar Impactor	1993
Kaguya [43]	65.5° S	80.4° E	Japan	Lunar Impactor	2009
Hakuto-R M1 (Intended site) [44]	47.5° N	44.4° E	Japan	Lunar Lander	2023
SLIM	13° S	25° E	Japan	Lunar Lander	2024
Chandrayaan-1 Moon Impact Probe [37]	89.9° S	122.9° W	India	Lunar Impactor	2008
Chandrayaan-2 Vikram Lander [38]	70.9° S	22.8° E	India	Lunar Lander	2019
Chandrayaan-3 Lander [39]	69° S	32° E	India	Lunar Lander	2023
Beresheet Lander [48]	32.6° N	19.3° E	Israel	Lunar Lander	2019
Ranger 4 [14]	15.5° S	130.7° W	U.S	Lunar Impactor	1962
Ranger 6 [18]	9.3° N	21.5° E	U.S	Lunar Impactor	1964
Ranger 7 [19]	10.6° S	20.7° W	U.S	Lunar Impactor	1964
Ranger 8 [20]	2.6° N	24.8° E	U.S	Lunar Impactor	1965
Ranger 9 [21]	12.8° S	2.4° W	U.S	Lunar Impactor	1965
Surveyor 1 [49]	2.5° S	43.3° W	U.S	Lunar Lander	1966
Surveyor 2 [50]	5.5° N	12.0° W	U.S	Lunar Lander	1966
Surveyor 3 [51]	3.0° S	23.4° W	U.S	Lunar Lander	1967
Surveyor 4 [52]	0.5° N	1.4° W	U.S	Lunar Lander	1967
Surveyor 5 [53]	1.4° N	23.2° E	U.S	Lunar Lander	1967
Surveyor 6 [54]	0.5° N	1.4° W	U.S	Lunar Lander	1967
Surveyor 7 [55]	41.0° S	11.4° W	U.S	Lunar Lander	1968
Apollo 11 [23]	0.7° N	24.5° E	U.S	Crewed Lunar Landing	1969
Apollo 12 [23]	3.2° S	23.5° W	U.S	Crewed Lunar Landing	1969
Apollo 14 [23]	3.7° S	17.5° W	U.S	Crewed Lunar Landing	1971
Apollo 15 [23]	26.1° N	3.6° E	U.S	Crewed Lunar Landing	1971
Apollo 16 [23]	9.0° S	15.5° E	U.S	Crewed Lunar Landing	1972
Apollo 16 [56]	1.92° N	4.16° W	U.S	S-IVB Booster Disposal	1972
Apollo 17 [23]	20.2° N	30.8° E	U.S	Crewed Lunar Landing	1972
Lunar Prospector [25]	88.1° S	45° E	U.S.	Lunar Impactor	1999
L-CROSS Centaur Probe [26]	85° S	43° W	U.S.	Lunar Impactor	2009
LADEE [29]	11.82° N	266.75° E	U.S	Lunar Orbiter	2014
IM-1 [57]	80° S	1° E	U.S.	Lunar Lander	2024
Luna 2 [58]	29.1° N	0° E	U.S.S.R	Lunar Impactor	1959
Luna 5 [59]	31.0° S	8.0° W	U.S.S.R	Lunar Lander	1965
Luna 7 [60]	9.0° N	49.0° W	U.S.S.R	Lunar Lander	1965
Luna 8 [61]	9.1° N	63.3° W	U.S.S.R	Lunar Lander	1965
Luna 9 [62]	7.1° N	64.4° W	U.S.S.R	Lunar Lander	1966
Luna 13 [63]	18.9° N	62.1° W	U.S.S.R	Lunar Lander	1966
Luna 16 [64]	0.5° S	56.4° E	U.S.S.R	Lunar Lander	1970
Luna 17 [65]	38.3° N	35.0° W	U.S.S.R	Lunar Lander	1970
Lunokhod 1 [65]	38.2° N	35.0° W	U.S.S.R	Lunar Rover	1970
Luna 18 [66]	3.6° N	56.5° E	U.S.S.R	Lunar Lander	1971
Luna 20 [67]	3.8° N	56.6° E	U.S.S.R	Lunar Lander	1972
Luna 21 [68]	25.9° N	30.5° E	U.S.S.R	Lunar Lander	1973

Table 1 (continued)

Mission name	Latitude	Longitude	Country	Purpose	Year
Lunokhod 2 [68]	5.9° N	30.5° E	U.S.S.R	Lunar Rover	1973
Luna 24 [69]	12.7° N	62.2° E	U.S.S.R	Lunar Lander	1976
Luna 25 [33]	57.86° S	61.36° E	Russia	Lunar Crash	2023

**Fig. 1** International Historical Sites: Lunar Midlands (Country Specific, Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)

satellite is currently orbiting the L2 Earth-Moon Lagrange point - another first for lunar missions [1].

The E.U. has flown only one lunar landing mission: an impactor vehicle named the Small Missions for Advanced Research in Technology (SMART-1). This mission was the E.U.'s first mission to the Moon and served as a technology demonstrator, specifically demonstrating solar-electric propulsion capabilities. SMART-1 launched in 2003 and became the first European spacecraft to successfully enter lunar orbit, and the vehicle impacted the Moon in 2006 [36]. Similar to the E.U., India's first lunar landing mission was also an impactor. The impactor, named the Moon Impact Probe, was carried and released by Chandrayaan-1, an orbiter designed to collect images and conduct remote sensing of the lunar surface. The Moon Impact Probe successfully completed its mission in November 2008 by impacting near the South Pole, thereby making India the fourth country to land an object on the Moon [37]. India launched Chandrayaan-2 in 2019 to attempt a soft-landing near the South Pole. Unfortunately, contact was lost with the lander while it descended, resulting in an unsuccessful landing. Estimates of where the lander impacted were refined over time and a specific location was eventually identified [38]. India's most

recent mission, Chandrayaan-3, provides yet another cite applicable to this atlas [39].

Israel conducted its first Moon mission in 2019 with the Beresheet lander, a commercial vehicle developed by the non-profit organization SpaceIL. Intended to land in the northern region of Mare Serenitatis, Beresheet suffered a systems failure during its descent to the lunar surface and crashed, destroying the vehicle [40, 41]. Next, Japan has completed two lunar impactor missions, with development for each managed by the Institute of Space and Astronautical Science, a precursor organization for Japan Aerospace Exploration Agency (JAXA) [41]. The first of these missions, Hiten, the first robotic lunar probe launched since 1976, was deliberately crashed into the Moon three years after launch in 1993 [42]. The second mission, Kaguya (formally known as SELENE, or the SELEnological and ENgineering Explorer), was intended to collect data on the Moon's origin, and it impacted the surface in 2009 [43]. A third Japanese mission, the Hakuto-R Mission 1, was a lunar lander operated by the private company iSpace that ran out of fuel during descent and crashed into the Moon. The intended landing site was in Atlas crater; however, the crash site has not yet been identified as communication after

landing was not established and surveys of potential crash sites are pending [44].

2.3 Map of Historical Lunar Sites

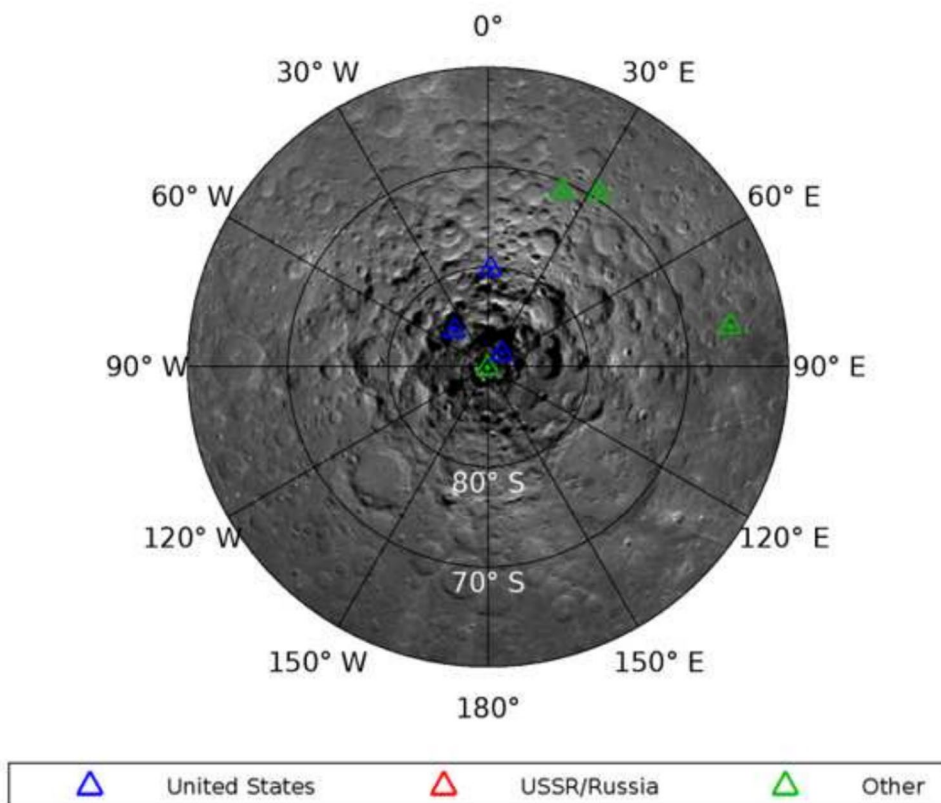
Table 1 presents all historical sites identified for the historical lunar site maps. Note that sites located in the South Pole region are shown in bold to differentiate them from sites located in the Midlands region. Figures 1 and 2 show the historical landing site maps, and the color of the marker indicates the country of origin with blue representing the U.S., red representing U.S.S.R./Russia, and green representing all other countries included as described previously. Without any historical landing sites in the North Pole region, a North Pole map is not included in this section. Also, Fig. 1 has a maximum latitude of 60° (both North and South). Those entries will be included in the Polar Region map for visual purposes.

Figures 1 and 2 indicate the majority of historical missions landed in the lunar Midlands region. This includes all of the U.S.S.R.'s missions, 14 in total. It is clear from Fig. 1 that areas such as Oceanus Procellarum, Mare Imbrium, and Mare Tranquillitatis were favored landing areas. All crewed lunar landings (Apollo program) occurred in one of these three areas except for Apollo 16, which landed in the relatively smooth Cayley Plains with a specific mission to

collect samples for studying the region's potential volcanic origins [70, 71]. Japan's Hiten, the first of their two lunar impactor missions, was released from a separate spacecraft performing a fly-by of the Moon and impacted in the lunar Midlands region [42]. Almost all Midland historical landing sites are located on the near side of the Moon (90° W to 90° E longitude) apart from Chang'e 4 and Ranger 4, a lunar impactor that impacted the far side of the Moon in 1962 [14]. Most soft-landing missions appear to have targeted areas of sufficient flat terrain with minimal gradient. Lunar impactor missions, or missions in which a soft-landing is not attempted, would not have required specific terrain to facilitate impact.

Seven historical landings have occurred in the lunar South Polar region. These missions were performed by China, India, Japan, and the U.S., with the first polar landing occurring in 1999. India's Chandrayaan-1 Moon Impact Probe impacted very close to the South Pole at a latitude of 89.9° S. This site was deliberately chosen with the probe transmitting data during its descent. Data from the Moon Impact Probe suggested there may exist water in the lunar atmosphere, but this finding is deemed inconclusive and is awaiting future verification [72]. India's second lunar probe mission, the Chandrayaan-2 Vikram Lander, crash-landed in 2019 after the Indian Space Research Organisation (ISRO) lost contact with the vehicle during descent [38]. A soft landing by the

Fig. 2 International historical sites: South Polar Region (Country Specific, Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)



ISRO with the Chandrayaan-3 mission is also listed [39]. Japan's second impactor, named Kaguya, impacted near the Gill Crater at 65.5° S longitude after completing its initial mission while in lunar orbit [43]. Finally, the U.S. has landed two probes in the lunar South Pole region—the Lunar Prospector and L-CROSS' Centaur Probe. The Lunar Prospector deliberately crashed into the lunar South Pole region after the satellite completed its original mission of collecting data to create maps of the lunar surface while orbiting the Moon [25]. With the second mission, the Centaur Probe separated from L-CROSS and impacted the lunar South Pole in 2009. Initial data collected from this impact indicated the presence of water at the South Pole. Other data from L-CROSS released in 2010 showed evidence of useful minerals in the lunar soil and that the Moon has an active water cycle [26].

3 Proposed Lunar Basing and Infrastructure

Data regarding lunar surface operations can also be derived from proposed works. Academic work proposing future lunar surface projects provide specific locations and support their work with justifications as to why certain sites are more viable than others. Common areas for proposed projects can be identified as many projects seek to achieve similar goals. These works frequently discuss proposed locations as site selection can heavily influence the mission's requirements and vice versa. The two main project types collected for this atlas are proposed bases and infrastructure. A base is defined as a site whose main mission is sustaining human life on the Moon; alternatively, infrastructure is defined as any other type of construction that may support the main base functions and/or mission(s). Examples of infrastructure include geographically separated stellar observatories, communications antennas, and mines. The resolution of information present within the consulted sources widely varied, and some sources only provided general regions for a project (i.e., a lunar pole), while others provided specific locations that included coordinates and in-depth discussion regarding the reasons for a specific location selection. All sources that provide any sort of location, including general regions, are included in this atlas in an attempt to be as expansive as possible for basing/infrastructure site selection. Table 2 outlines each location portrayed in Figs. 3 and 4, the proposed lunar basing and infrastructure maps for the Midlands and Polar regions, respectively.

A survey of historical and contemporary scientific literature reveals a commonality in location selection for proposed basing and infrastructure locations. The most prevalent general location proposed is one of the lunar poles. While some sources differentiated between the North and South Pole, with some even going as far to identify specific craters or other topographic features near one of the poles, many of

the advantages identified are shared between the North and South Poles [78, 88]. Overall, the poles can potentially provide sites of either perpetual darkness or constant illumination, although this occurrence remains to be fully proven [73, 78]. Some sources estimate that any structures built at one of the poles would be illuminated for at least 90% of the lunar orbit with a constant half of the sky continuously visible for observation [73, 90]. This, combined with areas of perpetual darkness may prove advantageous to the long integration times needed for deep space infrared observation [77]. Having illumination for 90% of the lunar orbit could also provide a source of power through solar arrays as a polar structure abbreviates the lunar night from fourteen days to three. In addition to the benefits of being illuminated, sources also claim that either pole would likely provide access to minerals of interest, including oxygen and hydrogen that could be derived from ice believed to be present at the poles [74]. Previous research has also pointed to the lunar poles as being likely locations of water on the Moon's surface [88]. As mentioned before, this has been tentatively proven by impactor missions launched over the past two decades [26]. Finally, a polar-located base also provides the possibility of quick access to and from any space station in a polar orbit. A space station in a lunar polar orbit would have a revisit time of around two hours, meaning there is a chance for launching to the station every 2 h [93].

Naturally there are some negatives to choosing a lunar pole as the location for a project. For example, the incident angle of sunlight at the lunar poles will leave everything in extreme contrast and decrease visibility. Another disadvantage is the difficulties in communicating with the Earth. Relays, likely positioned at the L1 or L2 Lagrange points or even in lunar orbit, would be needed to provide constant communications channel with Earth-based ground stations [73, 92]. Finally, while not an immediate issue, operating a human settlement on one of the lunar poles would likely produce significant electromagnetic noise. This noise would then affect any future projects on the far side of the Moon, an area that has been identified as having the benefit of being shielded from similar noise generated by Earth. Being relatively noise free, the lunar far-side has been identified as a good place for deep space observation. Mitigating noise generation could potentially be a future requirement of a polar base as lunar infrastructure develops [93].

Beginning with the South Pole, a specific benefit of this location is the ability of maintaining a continuous view of the southern celestial field (from the perspective of Earth). Although the North Pole also provides a continuous view of stars, the southern celestial field is less explored [93]. Some specific sites at the South Pole include the Shackleton Crater and Mons Malapert (Malapert Crater) [88, 89]. A site at the Shackleton Crater, specifically on the rim of the crater, would provide near constant solar illumination

Table 2 Proposed basing and infrastructure specifications

Region	Relative location	Latitude	Longitude	Country	Purpose
North Polar	North Pole [73]	90° N	0° E	U.S	Base
	North Pole [74]	90° N	0° E	U.S	Mining
	North Pole [75]	90° N	0° E	U.S	Observatory
	Whipple Crater [76]	89.1°	118.2° E	U.S	Base
	North Pole [77]	89.1° N	118.2° E	U.S	Observatory
	Peary Crater [78]	88.6° N	33° E	U.S	Base
	North Pole [79]	88.6° N	33° E	U.S	Base
Midlands	Mare Serenitatis [74]	21° N	29.5° E	U.S	Mining
	(Tarus-Littrow)				
	Mare Imbrium [80]	18.9° N	3.7° W	U.S	Base
	(North of Montes Apenninus)				
	Oceanus Procellarum [81]	18.5° N	63° W	U.S.S.R	Base
	Oceanus Procellarum [82]	18.4° N	57.4° W	U.S	Base
	Sinus Aestuum [80]	14.5° N	11.3° W	U.S	Base
	(near Erathosthenes Crater)				
	Oceanus Procellarum [83]	11.9° N	50.8° W	U.S	Base
	(Marius Hills)				
	Reiner Gamma [82]	7.5° N	59° W	U.S	Base
	Sinus Medii [84]	4.1° N	10.5° E	U.S	Base
	(Agrippa Crater)				
	Sinus Aestuum [80]	2.4° N	1.7° E	U.S	Base
	(near Sinus Medii)				
	Mare Smythii [85]	1.3° N	87.5° E	U.S	Base
	Grimaldi Crater [82]	5.5° S	68.3° W	U.S	Base
South Polar	Mare Orientale [86]	19.4° S	92.8° W	U.S	Base
	Tsiolkovsky Crater [87]	20.4° S	119.1° E	U.S	Radio Antenna
	Amundsen Rim [8]	83.5° S	72° E	U.S	Base
	Nobile Rim 2 [8]	84° S	61° E	U.S	Base
	Leibnitz Beta Plateau [8]	84.5° S	32° E	U.S	Base
	Mons Malapert [88]	84.9° S	12.9° E	Russia	Base
	Mons Malapert [82]	84.9° S	12.9° E	U.S	Base
	Nobile Rim 1 [8]	85° S	40° E	U.S	Base
	Shackleton Crater [89]	86° S	0° E	U.S	Base
	Malapert Massif [8]	86° S	1° W	U.S	Base
	de Gerlach-Kocher Massif [8]	86° S	122° W	U.S	Base
	Haworth [8]	86.5° S	30° W	U.S	Base
	de Gerlache Rim 2	87.8° S	67° W	U.S	Base
	de Gerlache Rim 1 [8]	88° S	65° W	U.S	Base
	Faustini Rim A [8]	88° S	90° E	U.S	Base
	Peak Near Shackleton [8]	89° S	120° E	U.S	Base
	Connecting Ridge [8]	89° S	150° W	U.S	Base
	Ridge Extension [8]	89° S	90° W	U.S	Base
	Shackleton Crater [90]	89.9° S	0° E	U.S	Base
	South Pole [4, 91]	90° S	0° E	China/Russia	Base
	South Pole [92, 93]	90° S	0° E	U.S	Base
	South Pole [74]	90° S	0° E	U.S	Mining
	South Pole [75]	90° S	0° E	U.S	Observatory

Fig. 3 Proposed basing/infrastructure locations in the Lunar Midlands. (Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)

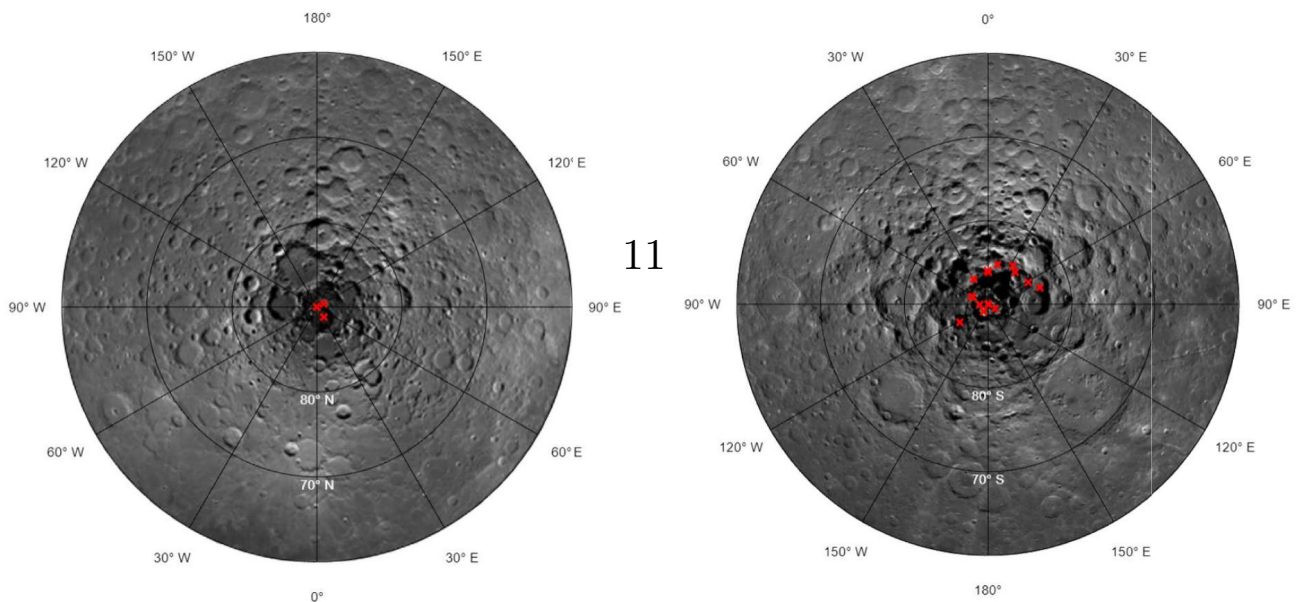
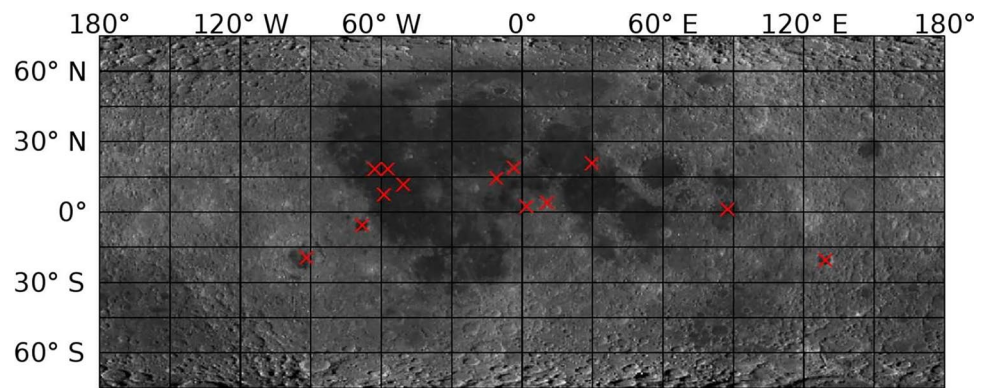


Fig. 4 Proposed basing/infrastructure locations in the (a) North Polar and (b) South Polar Regions. (Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)

and allow for constant Earth visibility [89]. Mons Malapert is also identified as a potential site, but construction there may prove difficult due to the presence of steep mountain slopes. If the slopes can be overcome during construction, then a basing project there may be able to take advantage of unique conditions afforded by the mountain slopes, including unique shade conditions and increased visibility over the rest of the crater [88]. Regarding the Artemis III potential sites, one of the main concerns is accessibility for traversing the lunar surface [8]. These sites are chosen with that in mind, making their integration essential for transport missions on the Moon, and in the case for this atlas, denote an important aspect for lunar basing and infrastructure. This is as important as any other feature when determining a location for infrastructure, as sites may only house hardware essential for their respective location. This also brings light to the networking between multiple sites, in that making

resources abundant between sites may pose a higher potential for growth and prosperity.

Some sources prefer the North Pole as a lunar base or infrastructure site, with one base construction project identifying Whipple Crater as an ideal site. This selection is due to its large, flat area that overlooks a deep cliff. It would also provide areas of perpetual darkness for use as sleeping quarters and cryogenic resource storage. Unfortunately, it would not likely be able to provide areas of continuous solar illumination [76]. Other sources view the North Pole as a location that could provide nearly constant temperature [91]. Other North Pole sites identified include Peary Crater and Polaris Crater, but these sites lacked specific reasons for their selection [78, 79].

Locations other than the lunar poles have been proposed as potential sites for both lunar bases and infrastructure. First, on the near-side Midlands region of the Moon, a base

located near the equator is continuously visible to Earth. This would make communications easier as a direct line-of-sight is always present, thus eliminating the immediate need for relay stations [84]. A challenge of building in this location is that the lunar night lasts approximately 14 days, an extended duration of darkness that must be factored into the design of any project located in this area [90].

Lunar *maria* are commonly identified as viable options for construction projects. A lunar *mare* is a large plain on the lunar surface that originated due to a combination of asteroid impacts and volcanic activity [94]. These sites are commonly deemed desirable due to their flatness, among other positive characteristics. Flatness is a benefit as it would make transport and construction easier [81]. One example is Oceanus Procellarum (Ocean of Storms), which is given the title of *oceanus* as it is the largest of all the lunar *maria* [95]. In addition to being flat, it is believed that there are natural resources present in Oceanus Procellarum [32]. In particular, the Soviets planned the Oceanus Procellarum as the location for not only their first human lunar expeditions, but also the construction of a Moon base. Conceptualized in 1967 with the intent of being built by the end of 1969, Moon base *Galaktika* feature a crew of between four and twelve cosmonauts, and consist of an astronomy laboratory, and a lunar rover capable of extending surface exploration up to 250 km from the base [32]. Two other bases, *Zvezda* and *Zvezda II*, were planned in the 1970s-1980s for construction in the Oceanus Procellarum. However, the plans for all three Moon base concepts were shelved due to the failure of the Soviet human exploration program and the deferment of resources to other space programs such as near-Earth space stations and lunar/planetary robotic exploration [32].

A specific location identified within the Oceanus Procellarum is the Marius Hills region. This is a location that has been studied as a potential location for lunar lava tubes to be present. These natural caverns, which are discussed later in more detail, are seen as a natural way to protect a lunar site from radiation and meteorites [83]. There is also geography of interest in this area, allowing for valuable scientific research to occur [81]. Mare Serenitatis has been determined to be a viable location for a mine due to a belief that there are deposits of resources of interest [74]. Mare Orientale is another site that is flat and is believed to hold valuable resources [86].

A unique mare that has been proposed for basing is Mare Smythii, a plain located on the lunar limb. This means that at this location, the Earth is sometimes visible and sometimes not as a result of the Moon's orbital motion—a phenomenon that can be both a benefit and disadvantage. When not in view of the Earth, this location is shielded from a significant amount of Earth's electromagnetic noise; when in view of the Earth, line-of-sight communications are enabled. These factors can be great benefits for many lunar projects, but they

are mutually exclusive and uncontrollable. Specifically, it is not possible to have both noise shielding and line-of-sight communications. Since the exact location of this site relative to Earth is dictated by the Moon's natural motion, any project built in this area would not have control over when it is in view of the Earth. This periodic phenomenon would have to be accounted for if planning a construction project in this area. Other benefits of Mare Smythii include easy access in minimum energy trajectories from near-Earth orbits, diverse terrain, and access to locations useful in characterizing the lunar global heat flow [85].

Apart from near-side locations, Tsiolkovsky Crater on the far-side of the Moon is identified as an ideal site for an observational radio antenna. An antenna built there would be separated from any near-side base and, as mentioned before, be shielded from electromagnetic noise coming from Earth. It also has a thick regolith, and interesting geological features for study. Negatives of this site include natural walls that rise above the lunar floor that would limit the visible portion of the sky. It is also a potential source of helium-3, an isotope that may be someday power reactors on Moon (or on Earth). Not many potential locations of this resource have been identified, meaning this area may eventually become a bustling mining region. However, mining sites would negate the benefits of Tsiolkovsky Crater being far from other basing/infrastructure sites and noise shielded [87].

A final general area commonly proposed as an advantageous location for a lunar base is inside of a lava tube. Lava tubes are natural conduits formed by previous basaltic lava flows in which lava channeled beneath the lunar surface and created tube-shaped passages [96]. A lava tube would provide many benefits to a lunar base project, including protection from radiation, access to unique geology, shielding from meteoroids, constant temperature, additional space for disposing of waste, and potentially resources of interest for mining [83, 97, 98]. Unfortunately, while the benefits of this location would be significant, the existence (and in turn location) of these lava tubes is unconfirmed [83, 98]. The conclusion that the Marius Hills region contains a lava tube comes from gravitational pattern analysis performed over a large hole in the surface that was identified at this location [83].

4 Combined Lunar Map

All of the coordinates for the historical sites and proposed basing locations were combined and plotted onto one set of maps to conclude this atlas. Such a combined map provides the best opportunity to identify areas in which there is agreement of data from multiple sources spanning different decades and genres. It can be argued that the coordinates identified in this paper represent the likely locations of future human lunar exploration and settlement, particularly those

locations cited by the scientific community. Although large expanses of the lunar surface do not feature either historical sites or proposed basing locations, this does not mean that at some point in the future that human activity may extend into these regions following the establishment of basing (and infrastructure) in areas featuring minimal gradient, advantageous solar illumination, and/or desirable conditions for a given mission. In addition, the absence of any historical or proposed activity in a given region may also be a function of a lack of sufficient information regarding the mission utility and a yet-to-be-defined case for pursuing a landing at and development of the region. The combined map for the lunar Midlands is shown in Fig. 5, while the North and South Polar regions are shown in Fig. 6. A heat map illustrating the lunar site count is also provided in Fig. 7 depicting the frequency of sites across the lunar surface.

After reviewing the combined maps, there are several examples of agreement and disagreement between the

different data sources consulted. Beginning with the Midlands region, proposed basing sites frequently appear to be in the same areas as astroarchaeology sites. This co-location is likely because these areas, such as Oceanus Procellarum and other lunar maria, are level. Less mountainous regions makes it easier to land and erect infrastructure, so it therefore is logical for these areas to have been targeted in the past and continue to be targeted in proposed works. Other proposed sites that do not align with astroarchaeology sites are located in either other maria or the interior of craters. Overall, the vast majority of historical and proposed sites for the Midlands region are on the near side of the Moon. This trend is likely a result of ease of landing, surface operations, and communications with Earth.

Looking next at the North Polar region, there are only proposed sites present. All proposed basing sites are located on or very close to the North Pole, which is not surprising due to the unique solar illumination and other conditions

Fig. 5 Combined map for the Lunar Midlands [99]. (Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)

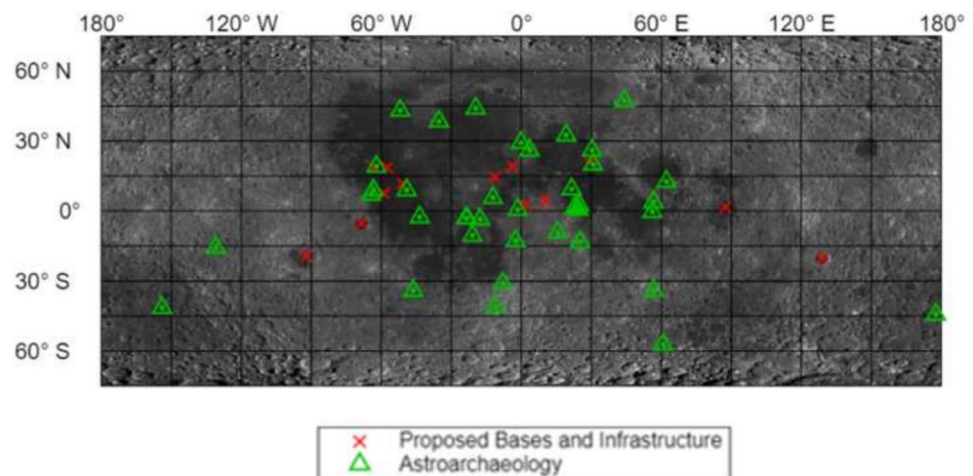


Fig. 6 Combined map for the (a) North Polar and (b) South Polar Regions [99]. (Imagery adapted with permission from Lunar QuickMap, a collaboration between NASA, Arizona State University, and Applied Coherent Technology Corp. Copyright 2023)

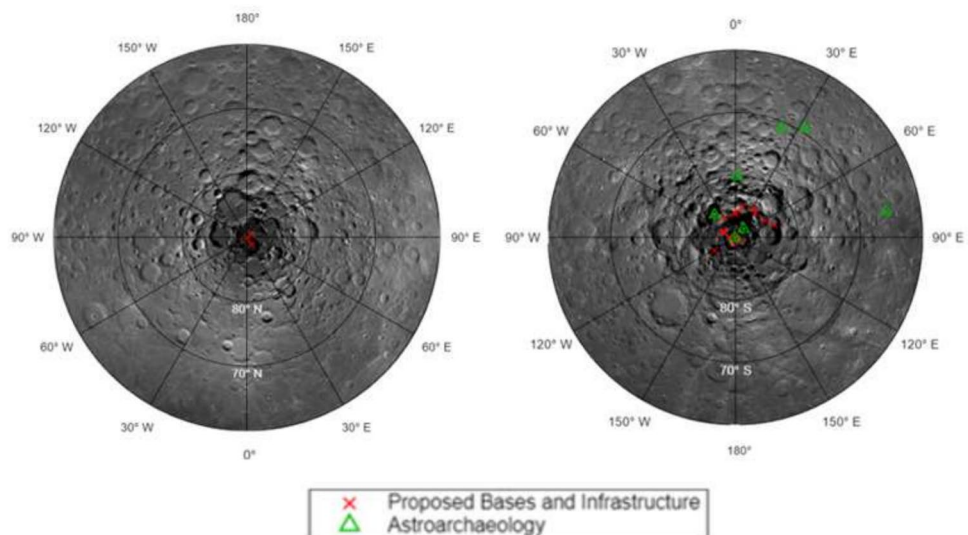
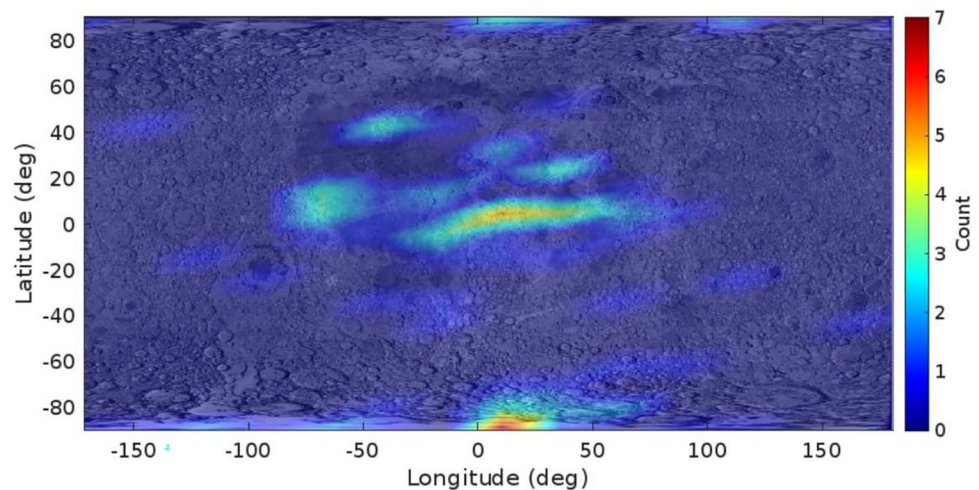


Fig. 7 Heat map of historical and proposed lunar sites



afforded by this area. These sites offer the same advantages as what is offered by other maria and crater sites closer to the equator on the near side of the Moon. However, being closer to the poles would result in diminishing some of the advantages of being closer to the equator (i.e., being easily and constantly visible from Earth). With no assessed advantage for moving away from the equator or the immediate vicinity of the pole into the mid-latitude region, there is no reason to plan a mission in this area due to the challenges associated with solar illumination and communication. This is reflected in the absence of any proposed basing/infrastructure in the mid-latitude region.

The South Polar region shows similar results to the North Polar region, but with a larger amount of historical sites as illustrated by the stereographic map in Fig. 6 and the heat map in Fig. 7. Once again, all proposed basing sites are located on or very close to the South Pole. Figure 6 shows that the historical sites are quite spread out and cover a wide spread of latitude. With multiple missions from multiple countries that specifically targeted the South Pole or an area relatively close to it, it can be concluded that there is likely more interest in sending missions to the South Pole than the North Pole. This could point towards the South Pole as being favored for a future polar base due to not only the environmental conditions of this region, but also the mission/spaceflight experience accrued travelling to this region rather than the North Pole.

5 Lunar Infrastructure and Space Debris Considerations

As lunar infrastructure is developed and expands, risk associated with debris fields becomes a major aspect of location choice. As of now (and shown previously in this atlas), there are instances in which, for example, a lunar mission

is designed to impact the Moon's surface for end-of-life disposal. Other sources of manmade hazards to the lunar surface result from catastrophic spacecraft breakup events that result in the generation of a debris cloud in low lunar orbit (LLO). The likely impact locations of entire vehicles or the debris from breakup events are important aspects for determining locations of infrastructure due to the increased risk in caused by such impacts with damaging critical infrastructure and associated equipment. Dust from said impacts may also result in a generation of compounds/materials (e.g., quartz) in the exosphere that can potentially damage essential systems, such as solar panels. In this section, a comparison between simulated debris impact locations and proposed basing/infrastructure locations will be shown in order to give further understanding of the risk associated with these locations and provide further insight into factors that should go into establishing a site location.

To be brief, this atlas considers a sample set of lunar debris generated via Monte Carlo campaigns by Boone and Bettinger [100] associated with spacecraft breakup events in LLO. First, some preliminary information must be included to describe the integration of lunar debris with location selection. The Monte Carlo simulations investigated the propagation of artificial debris across all orbit types in LLO [100]. A set of 2000 initial breakup events at random initial states, resulting in 200,000 particles propagated via high-fidelity trajectory modeling for in LLO for a maximum duration of 5 years. All particles impacting the lunar surface were then graphed via color map based on impact count to illustrate impact concentration as a function of color. The resulting lunar impact regions of the debris fragments were then compared with the lunar activity atlas. One of the products of the Monte Carlo simulations was a heat map illustrating the impact locations of all associated fragments. This heat map is shown along with the astroarchaeological and proposed bases/infrastructure locations in Fig. 8 [100]. This

Fig. 8 Debris impact for generic orbits in relation to historical and proposed infrastructure sites [100]

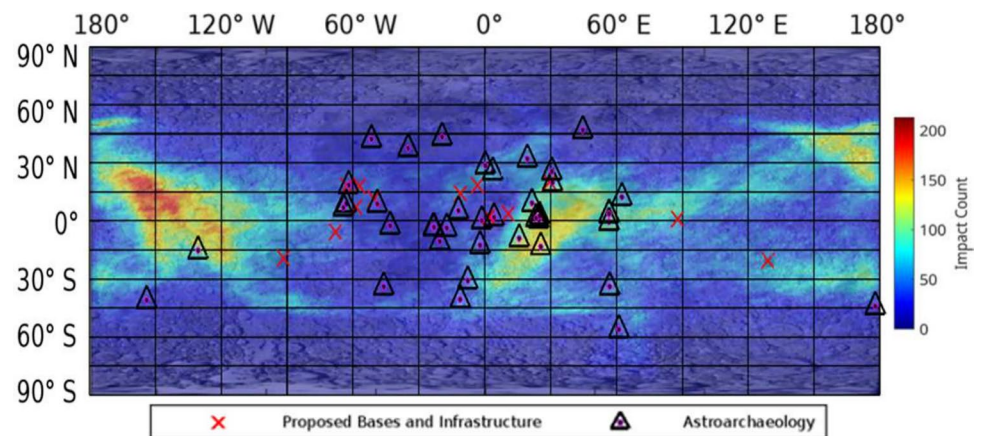


figure combines the location map from Fig. 5 with the debris impact data from [100] in a singular cartographical representation. Figure 8 provides an avenue for determining high-risk lunar locations for basing and infrastructure within the lunar midlands. This essential tool characterizes where long-term infrastructure should not be located to avoid debris impacts from breakup events in LLO.

Another set of Monte Carlo simulations addressed surface impact locations related to breakup events in orbits with inclinations between 80° and 100°, correlating to polar and near-polar orbits [100]. The debris field in Fig. 9 shows the resulting debris from these specific orbital trajectories from the five-year simulation runs, and contain all astroarchaeological and proposed basing/infrastructure locations for comparison.

If a location within the lunar midlands is a predetermined requirement for a basing and/or infrastructure location, then Fig. 8 is essential for understanding potential LLO debris

impact risks. The region spanning from 100° W to 0° correlates to a zone with the least risk from debris impact and damage, which encompasses sites such as Oceanus Procellarum, Reiner Gamma, and Grimaldi Crater. Uniquely, this zone also contains many astroarchaeology sites. Figure 9 can be used for determining risk of polar sites. One zone in particular, that spanning 40° E to 80° E and from 20° N to 85° S, contains a majority of the debris impact locations. These regions should be avoided when determining infrastructure locations, unless for a specific requirement and special shielding precautions considered for construction.

6 Conclusion

Historically, seven countries have ventured to the Moon (eight if separating U.S.S.R and Russia), with missions intended to either impact or soft-land on the lunar surface.

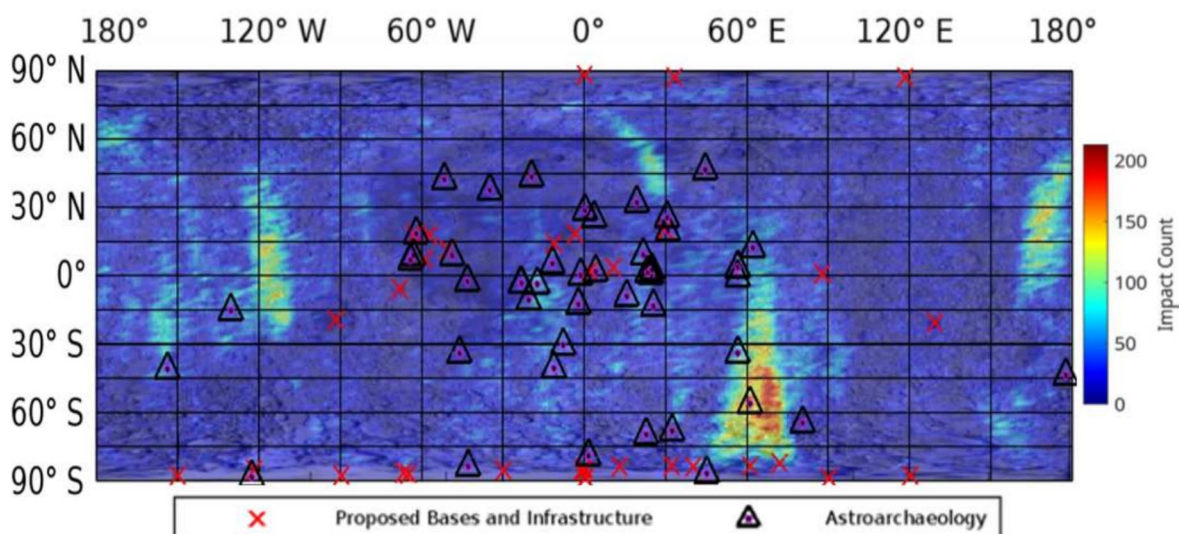


Fig. 9 Debris impact for near-polar orbiting vehicles (80° to 100° Inclination) in relation to all historical and proposed infrastructure sites [100]

Of the countries involved, the U.S. and U.S.S.R. pursued the most aggressive campaign of lunar exploration and have generated the highest number of landing sites for either crewed or uncrewed exploration. Most historical sites reside areas of flat terrain, thus demonstrating that low gradient was likely prioritized to make the landing process safer and easier. Some lunar impactor missions did not follow this trend, but this is logical as they followed an entirely different mission profile as their soft-landing counterpart missions. To date, only one mission has soft-landed on the far-side of the Moon (Chang'e 4), with all other activity being associated with impactor vehicles. Proposed base and infrastructure sites tend to fall into one of two general options. The first option is to select a basing site very close to one of the lunar poles as demonstrated by plans formulated by the Artemis program, as well as China and Russia. The poles provide some unique benefits such as unchanging views of space that could facilitate observation, as well as potential areas of constant solar illumination and darkness. The second option is to select in a maria or crater near the equator on the near-side of the Moon. These sites have the benefit of being continuously visible from Earth that would enable an easier and nearly continuous communications link.

This paper constitutes the first atlas of human activity on the lunar surface, with 92 annotated sites collected from historical lunar missions, proposed future basing and infrastructure reports. The collected sites are presented using a few basic maps of the lunar surface leveraging Mercator and stereographic projections, coupled with data tables outlining specific latitudinal and longitudinal coordinates for each site. This atlas also provides a risk assessment in relation to proposed basing locations, in which debris maps are compared to historical sites and proposed basing locations. Aside from potential uses with informing the site selection of a variety of future lunar exploration projects, this atlas serves to document the geographic regions of interest on the Moon from a historical and scientific perspective and should represent a living document to enable the continued chronicling of human lunar surface activity.

Author Contributions All authors reviewed the manuscript. R.B. initiated the manuscript idea, led the analysis, and supervised the composition of the manuscript. J.M. and T.G. conducted the majority of the manuscript drafting, with A.P. also contributing to the drafting. T.G. and A.P. created the initial figures and tables, but J.M. updated them for readability and new data from 2023–2024. N.B. contributed the debris information in Sect. 5.

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Data Availability The datasets generated for the current study are available following the submission and approval of a formal institutional request to the Air Force Institute of Technology (AFIT).

Declarations

Conflict of interest The authors declare no Conflict of interest.

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